SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

(Deemed to be University) • Chennai

ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO2 PROBLEM SOLVING ASSIGNMENT

Standards-Compliant Optical Fiber Link Design: From Communication Requirement to Deployment Decision

60 km · 1550 nm · 10 Gb/s with 100 Gb/s upgrade provision

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STAGE 1 – Interpretation of the Communication Requirement

Given system: 8 channels × 25 Gb/s NRZ = 200 Gb/s aggregate, 100 GHz (≈0.8 nm) spacing near 1550 nm, six amplified 80 km SSMF spans (480 km total).

- **Why the source matters:** 25 Gb/s NRZ needs ≈17.5 GHz of modulation bandwidth ($B \approx 0.7R_b$), so the source/modulator must respond at ≥ 20 GHz. A narrow linewidth (MHz-class) keeps each channel inside its 100 GHz slot and limits chromatic-dispersion broadening over 480 km. High output power is needed to overcome coupling, modulator and multiplexer losses and to keep OSNR high after six EDFAs.
- **Source characteristics:** bandgap E_g sets wavelength ($\lambda \approx 1.24/E_g$); internal quantum efficiency $\eta_{int} = \tau_{nr}/(\tau_r+\tau_{nr})$; threshold current I_{th} and slope efficiency η_s set $P = \eta_s(I-I_{th})$; linewidth and side-mode suppression ratio (SMSR) set spectral purity.
- **Modulation and temperature quantities:** relaxation-oscillation frequency f_r, 3 dB bandwidth, extinction ratio, chirp (linewidth-enhancement factor α_H), and wavelength-temperature coefficient (0.1 nm/°C).
- **Nonlinear quantities:** effective length L_{eff}, γ, P, SPM phase φ_{NL}, XPM/FWM efficiency (channel spacing and dispersion), and the SBS and SRS threshold powers (total power, linewidth).
- **Standards:** ITU-T G.957 (SDH optical interfaces: mean launched power, extinction ratio, SMSR, spectral width), IEEE 802.3bm/bs (100G/200G/400G Ethernet PHYs: OMA, TDECQ, SMSR, RIN, eye mask) and ANSI Z136 (laser eye safety: accessible emission limit for the total power on the fiber).
- **Compensation and integration:** D, β_2, γ, launch power and span count decide whether solitons or digital backpropagation (DBP) are sensible; footprint, power consumption, coupling loss and scalability decide discrete vs InP vs Si-photonics.

Problem Requirement Statement: The transmitter must deliver eight wavelength-locked (±0.1 nm), single-longitudinal-mode channels on the 100 GHz grid near 1550 nm, each carrying 25 Gb/s NRZ with ≥ 17.5 GHz bandwidth, low chirp, SMSR ≥ 30 dB and about 2 mW (3 dBm) per channel launch, while keeping SPM/XPM/FWM penalties small, SBS controlled, total launch power eye-safe, and satisfying G.957 / IEEE 802.3bm/bs, in a compact and reliable (preferably integrated) form.

STAGE 2 – Optical Source Selection

Recombination and bandgap. Electron–hole pairs recombine radiatively (photon emission, lifetime τ_r) and non-radiatively (traps, Auger; lifetime τ_{nr}). The photon energy is $\approx E_g$.

$\lambda = 1.24 / E_g = 1.24 / 0.80 = 1.55\ \mu m =$ ***1550 nm*** *(inside the 1550 nm window)*

$1/\tau = 1/\tau_r + 1/\tau_{nr} = 1/10 + 1/40 = 0.125\ ns^{-1} \rightarrow \tau = 8\ ns$

$\eta_{int} = \tau / \tau_r = \tau_{nr}/(\tau_r+\tau_{nr}) = 40/50 =$ ***0.80 (80 %)***

Threshold condition. Oscillation starts when round-trip gain equals loss: $\Gamma g_{th} = \alpha_{int} + (1/2L)\ln(1/R_1R_2)$, with phase condition $2\beta L = 2\pi m$. In a DFB/DBR laser the grating replaces the cleaved-facet feedback and selects a single mode. Above I_{th} stimulated emission dominates and a narrow line with high SMSR is produced. A multimode FP laser spreads power over several modes that walk through dispersion (mode-partition noise) and fall in neighbouring channels; a ≥ 30 dB SMSR with MHz linewidth keeps each channel well inside its 100 GHz slot and cuts inter-channel crosstalk.

Comparison of candidate sources (typical values for 1550 nm devices)

Source	Output power	Spectral width	Mod. speed	Efficiency / threshold	Fiber coupling
LED	≤ 0.1 mW	50–100 nm	< 1 GHz	Low; no threshold	Poor (broad beam)
FP laser	5–10 mW	1–3 nm (multimode)	≈ 10 GHz	Good; 10–30 mA	Moderate
DBR laser	5–20 mW	MHz-class, SMSR > 35 dB	≈ 10 GHz	Good; mode-hop risk, tuning section	Good
DFB laser	5–20 mW	1–10 MHz, SMSR > 40 dB	10–25+ GHz	Good; 10–20 mA	Good
VCSEL (1550)	1–3 mW	Single mode, tens of MHz	10–25 GHz	Very low threshold (< 5 mA); low power	Excellent (circular beam)
QD laser	5–10 mW	Narrow; low α_H	10–20 GHz (emerging)	Low threshold, temperature-insensitive	Good

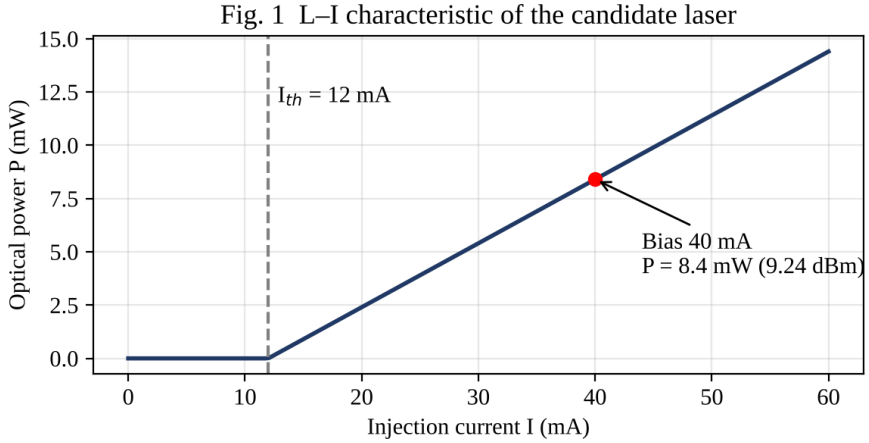
DESIGN DECISION 1: I recommend a **DFB (distributed-feedback) laser, operated as a CW source with an external electro-absorption modulator (EML)** optical source because it (1) emits at $\lambda = 1.55$ µm, (2) has $\eta_{int} = 80$ % and $I_{th} = 12$ mA, (3) delivers 8.4 mW (9.24 dBm) at 40 mA, (4) has a single longitudinal mode with SMSR > 35–40 dB and MHz-class linewidth for the 100 GHz grid, and (5) has a 20 GHz bandwidth, above the 17.5 GHz required.

Credible alternative: the **QD laser** becomes preferable if the link must run *uncooled* or at high case temperature (its threshold and wavelength vary far less with temperature), or when very low chirp and low RIN are required and the technology is mature enough for production.

STAGE 3 – Modulation, Chirp and Spectral-Stability Strategy

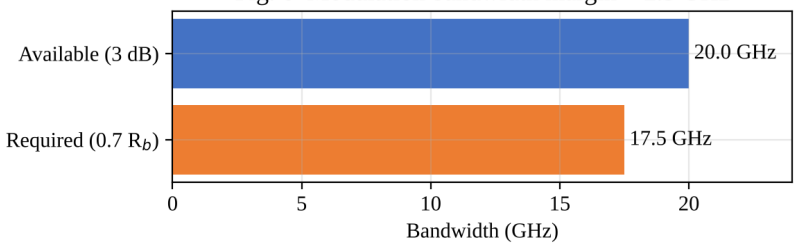
Output power: *$P = \eta_s(I - I_{th}) = 0.3$ mW/mA × (40 − 12) mA =* ***8.4 mW****; $P(dBm) = 10 \log_{10}(8.4)$ =* ***9.24 dBm****.*

Fig. 1 L–I characteristic of the candidate laser



Bandwidth: $B \approx 0.7R_b = 0.7 \times 25 =$ **17.5 GHz**. Available = 20 GHz ≥ 17.5 GHz, so it is **sufficient** with margin = 20 − 17.5 = **2.5 GHz (12.5 % of available)**. The margin is modest, so the driver, package and EAM must not add significant roll-off.

Fig. 6 Modulation bandwidth margin = 2.5 GHz



Rate equations (N = carrier density, S = photon density):

$$dN/dt = I/(qV) - N/\tau - v_g\, a\, (N - N_0)\, S$$

$$dS/dt = \Gamma\, v_g\, a\, (N - N_0)\, S - S/\tau_p + \Gamma\, \beta_{sp}\, N/\tau$$

Relaxation-oscillation frequency: $f_r = (1/2\pi)\ \sqrt{[\ v_g\, a\, S_0 / \tau_p\ ]} \quad \sqrt{(I - I_{th})}$, and $f_{3dB} \approx 1.55\, f_r$. For 20 GHz this implies $f_r \approx 13$ GHz. In **small-signal** modulation the response is linear with a resonance peak at f_r. In **large-signal** NRZ switching the laser shows turn-on delay, ringing at f_r, pattern-dependent jitter and large transient chirp, so the usable rate is lower than the small-signal bandwidth suggests.

Direct vs external modulation

Aspect	Direct modulation (DML)	External modulation (EAM/MZM)
Chirp	Large ($\alpha_H \approx 3$–5; transient + adiabatic chirp)	Small ($\alpha \approx 0$–1); EAM/MZM nearly chirp-free
Bandwidth	Limited by f_r (≈ 20 GHz here), sets ER/eye closure	EAM > 30 GHz; laser kept CW
Dispersion interaction	Chirp + anomalous D broadens pulse; reach only a few km at 25 Gb/s	Dispersion tolerance set by NRZ spectrum only
Distance	80 km without compensation	Unchirped NRZ ≈ 12 km uncompensated; with per-span DCM suits 6 × 80 km
Cost / power	Lowest, simplest	Higher (modulator + driver), but integrable in an EML

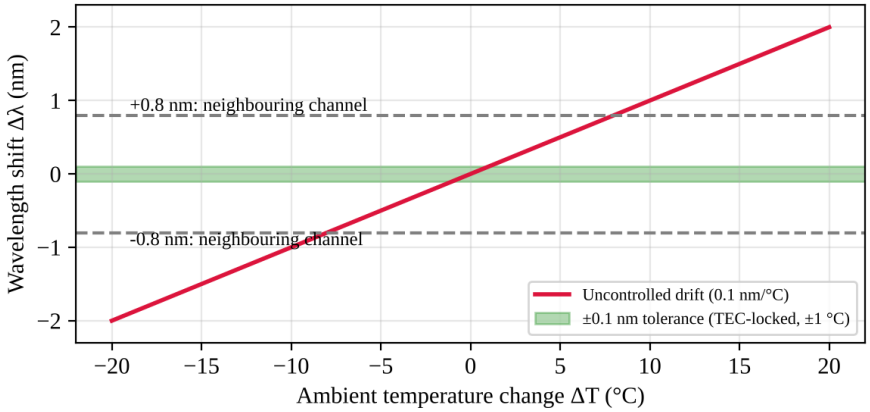
The unchirped dispersion limit was estimated with $L \approx c/(D\lambda^2 B^2)$ using D = 17 ps/(nm·km) (assumed), B = 25 GHz: L ≈ 12 km. Accumulated dispersion is 17 × 80 = 1360 ps/nm per span (8160 ps/nm over six spans), so per-span dispersion compensation (DCM/FBG) or electronic compensation is required.

Temperature drift

$\Delta\lambda$ = 0.1 nm/°C × 40 °C = ***4.0 nm*** *= 5 × 0.8 nm channel spacing → an uncontrolled laser would slip across five neighbouring channels.*

Required stability: $\Delta T = \Delta\lambda_{tol} / (d\lambda/dT)$ = 0.1 nm / 0.1 nm/°C = ***±1 °C*** *(2 °C window). A TEC loop with ±0.5 °C (±0.05 nm, 2× margin) is adopted.*

Fig. 2 Wavelength drift versus temperature



Method	Tuning range	Side effects	Verdict
TEC (temperature)	Several nm; correct for the full 4 nm drift	Slow (ms–s), uses power	Primary coarse control
Injection current	≈ 0.1–0.3 nm only (≈ 0.01 nm/mA, typical)	Changes power, chirp, ER	Only fine trim via wavelength-locker error signal

DESIGN DECISION 2: I recommend **external (EAM) modulation** and **TEC temperature control with an etalon wavelength locker (current used only for trimming)** because the laser gives 8.4 mW (9.24 dBm), the 20 GHz bandwidth exceeds the 17.5 GHz requirement by 2.5 GHz, the uncontrolled 4.0 nm drift (5 channel spacings) is held to ±0.05 nm by the ±0.5 °C TEC, and the low-chirp EAM avoids the chirp-dispersion penalty over 480 km.

STAGE 4 – Nonlinearity Investigation and Control

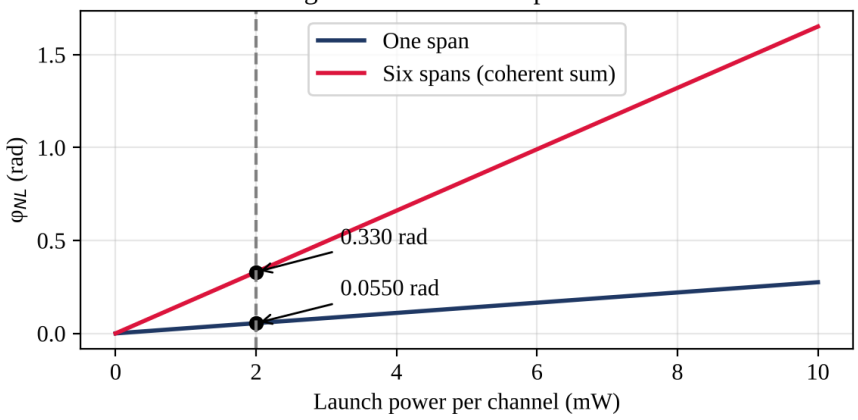
$\alpha = 0.2$ dB/km → $\alpha = 0.2/4.343 = 0.04605\ km^{-1}$; $\alpha L = 3.684$

$L_{eff} = (1 - e^{-\alpha L})/\alpha = (1 - e^{-3.684})/0.04605 =$ ***21.17 km***

φ_{NL} (per span) $= \gamma P L_{eff} = 1.3\ W^{-1}km^{-1} \times 0.002\ W \times 21.17\ km =$ ***0.0550 rad***

φ_{NL} (six spans, worst-case sum) $= 6 \times 0.0550 =$ ***0.330 rad*** *(< ≈1 rad guideline, assumed) → SPM is small for a 25 Gb/s NRZ signal.*

Fig. 3 SPM nonlinear phase shift



XPM and FWM. XPM: the intensity of one channel changes the refractive index seen by its neighbours, giving phase modulation that dispersion converts to intensity noise and timing jitter; it scales with P_{other} and with the walk-off between channels (small for 100 GHz spacing and large dispersion). FWM: channels at f_i, f_j, f_k generate $f_i+f_j-f_k$, producing crosstalk inside other channels;

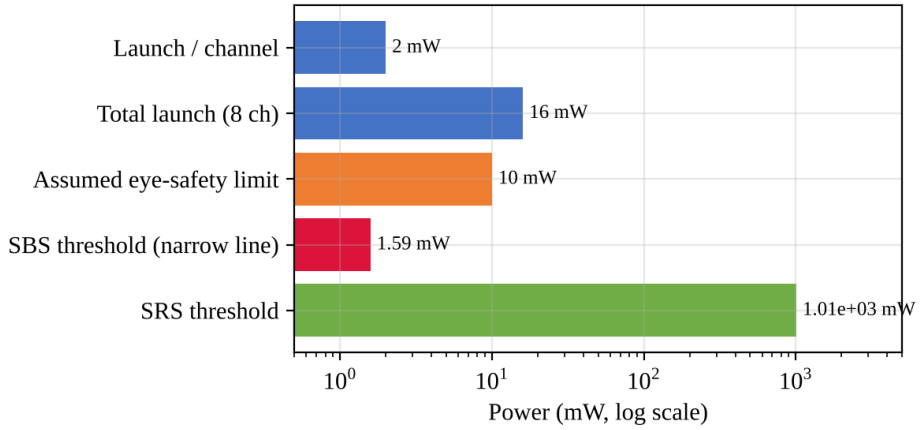
efficiency $\eta_{FWM} = \alpha^2/(\alpha^2+\Delta\beta^2)$ falls with channel spacing and dispersion, and the power grows as P^3. Estimate (D = 17 ps/nm/km, Δf = 100 GHz): $\Delta\beta = 2\pi\lambda^2 D\Delta f^2/c$ = 8.55 rad/km, $\eta_{FWM} \approx$ 2.9e-05, $P_{FWM} = \eta(6\gamma L_{eff})^2 P^3 \approx$ 6.3 nW (≈ -55 dB below one channel) per span. Equally spaced channels and large local dispersion in SSMF keep FWM and XPM small; they would worsen on dispersion-shifted fiber or with higher power.

SBS and SRS thresholds

$P_{th,SBS} \approx 21\, A_{eff}/(g_B\, L_{eff}) = 21 \times 80\times10^{-12} / (5\times10^{-11} \times 21169) =$ ***1.59 mW (2.0 dBm)***

$P_{th,SRS} \approx 16\, A_{eff}/(g_R\, L_{eff}) = 16 \times 80\times10^{-12} / (6\times10^{-14} \times 21169) =$ ***1.01 W*** ($g_R = 6\times10^{-14}$ *m/W assumed, as printed value is unclear)*

Fig. 4 Launch power versus SBS, SRS and safety limits



Comparison: SBS threshold (1.59 mW) is **below** the 2 mW per-channel launch for a narrow-linewidth CW line, so SBS must be controlled. SRS threshold (1.01 W) is ≈ 60× above the 16 mW total, so SRS is negligible.

Property	SBS	SRS
Origin	Photon–acoustic phonon scattering	Photon–optical phonon scattering
Gain bandwidth	≈ 20–100 MHz (narrow)	≈ 10–15 THz (broad)
Power transfer	Backward (reflected) – limits forward power	Forward; short-λ channels pump long-λ channels (tilt)
Effect on DWDM	Per-channel; saturates launch power, adds noise	Total power/bandwidth dependent; inter-channel crosstalk
Linewidth effect	Threshold rises when signal linewidth Brillouin bandwidth	Practically none

The 25 Gb/s NRZ data spectrum (tens of GHz) is far wider than the Brillouin gain bandwidth, and only the residual carrier component stays narrow; so external modulation raises the effective SBS threshold. Adding a small phase/frequency dither further broadens the line, giving a margin without reducing power.

Eye safety (ANSI Z136): Assumed limit: Class 1 accessible emission ≈ 10 mW (+10 dBm) at 1550 nm (order-of-magnitude assumption; Class 3R ≈ 5× higher, to be verified against the applicable standard). Total launch = 8 × 2 mW = 16 mW = 12.0 dBm → **exceeds** the assumed Class 1 limit, while each individual channel (2 mW) does not. The aggregate booster output must therefore be treated as a higher-class port with automatic laser shutdown (ALS/APR), connector caps and warning labels.

Two practical methods to reduce nonlinear penalty without losing signal level: (1) **external modulation + SBS dither/linewidth broadening** with unchanged 2 mW per channel; (2) **dispersion-managed span design (large-A_{eff} fiber and optimized per-span DCM/launch profile)** and keeping power only as high as OSNR requires (≈ 2 mW). Using low-noise EDFAs also lets the power be kept low.

DESIGN DECISION 3: I recommend a **launch-power-limited strategy with 2 mW/channel, SBS dither, external modulation, per-span dispersion management and ALS eye-safety shutdown** because φ_{NL} is only 0.330 rad over six spans, FWM/XPM are suppressed by the 100 GHz spacing and 17 ps/nm/km dispersion, SRS threshold (1.0 W) is 16 mW, and the SBS threshold (1.59 mW) is the only limit that is exceeded and is overcome by linewidth broadening.

STAGE 5 – Soliton Transmission, Digital Backpropagation and Integration

Soliton physics. In the anomalous-dispersion region ($\beta_2 < 0$) dispersion-induced chirp is opposite to the Kerr (SPM) chirp. For a sech pulse of the correct peak power they cancel exactly ($L_D = L_{NL}$, N = 1) and the pulse keeps its shape.

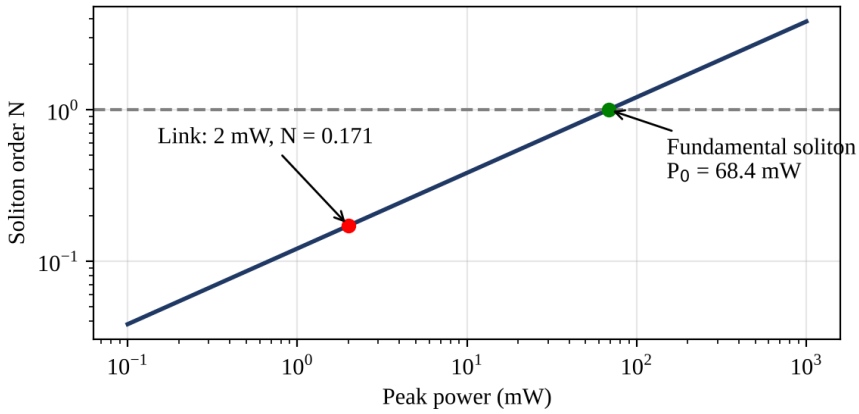
$L_D = T_0^2/|\beta_2| = (15\ ps)^2 / 20\ ps^2/km = 225/20 =$ ***11.25 km***

$L_{NL} = L_D \rightarrow 1/(\gamma P_0) = L_D \rightarrow P_0 = 1/(\gamma L_D) = 1/(1.3 \times 11.25) = 0.0684\ W =$ ***68.4 mW***

At 2 mW: $L_{NL} = 1/(\gamma P) = 1/(1.3 \times 0.002) =$ ***384.6 km***

$N^2 = L_D/L_{NL} = 11.25/384.6 = 0.0293 \rightarrow N =$ ***0.171***

Fig. 5 Soliton order versus peak power (T_0 = 15 ps)



Soliton feasibility: NOT feasible. N = 0.17 1 (dispersion dominates), so the pulse would broaden. A fundamental soliton would need 68.4 mW per channel, ≈ 34× the 2 mW launch; eight such channels = 547 mW, which would exceed the SBS and the assumed eye-safety limits. Fiber loss, amplifier ASE (Gordon–Haus jitter) and WDM soliton collisions further prevent it. Conventional NRZ with dispersion management is therefore chosen.

Digital backpropagation (DBP). The receiver (or transmitter) DSP numerically propagates the field backwards through a model of the link (inverse dispersion + inverse Kerr phase step by step), so deterministic dispersion and SPM/XPM are removed. **Limitation:** it needs the complete optical field (coherent detection with ADC/DSP) and, for the 8-channel link, all channels must be processed jointly (very high complexity/power); PMD and signal–ASE nonlinear noise cannot be compensated. **Verdict:** not appropriate as the primary method for this IM-DD 25 Gb/s NRZ link (SPM = 0.33 rad is small); single-channel or full-field DBP is reserved for a possible coherent upgrade.

Standards and verification. ITU-T G.957: mean launched power, extinction ratio, central wavelength, spectral width/SMSR, eye mask. IEEE 802.3bm/bs (25.78125 GBd lanes): OMA,

TDECQ/TDP, ER, SMSR, RIN, wavelength range, eye mask, jitter. Checking: power meter (mean power), optical spectrum analyzer (wavelength, SMSR, 100 GHz grid), sampling oscilloscope with eye mask (ER, TDECQ), BER tester (PRBS31) after 480 km (or emulator), and laser-safety classification test of the aggregate port.

Aspect	Discrete	Monolithic InP	Si-photonics + hybrid laser
Footprint	Large (8 TOSAs + mux)	Smallest for III-V PIC array	Small; large for the laser bonding
Power consumption	Highest (separate TECs/drivers)	Low (shared TEC, short paths)	Low-moderate (modulator + laser)
Coupling loss	Many couplings (≈ 1–3 dB each)	On-chip; one fiber/AWG interface	Fiber-chip coupling plus laser-chip coupling
Reliability	Proven but many joints	High (fewer joints), mature at 1550	Good, but hybrid bonding yield is critical
Scalability	Poor	Good (EML arrays, AWG on-chip)	Best for high volume and CMOS co-integration

Integration shortens the optical paths and reduces coupling loss, cuts size/power, and improves channel-to-channel wavelength matching, which helps meet the ±0.1 nm limit and lets EAM bandwidth exceed 20 GHz with low parasitics, although thermal crosstalk between neighbouring lasers must be managed.

DESIGN DECISION 4: I recommend a **monolithic InP 8-channel EML array (DFB + EAM) with an integrated AWG multiplexer, with receiver-side DSP/DBP left as an optional coherent upgrade** transmitter/integration strategy because 1550 nm InP gives the DFB + EAM on one chip with low coupling loss and small footprint; the system shows SPM of 0.33 rad (DBP not needed), soliton operation is infeasible (N = 0.17) and the standards (G.957, IEEE 802.3bm/bs) can be verified on one module. Si-photonics with a hybrid laser is the alternative for volume production or CMOS integration.

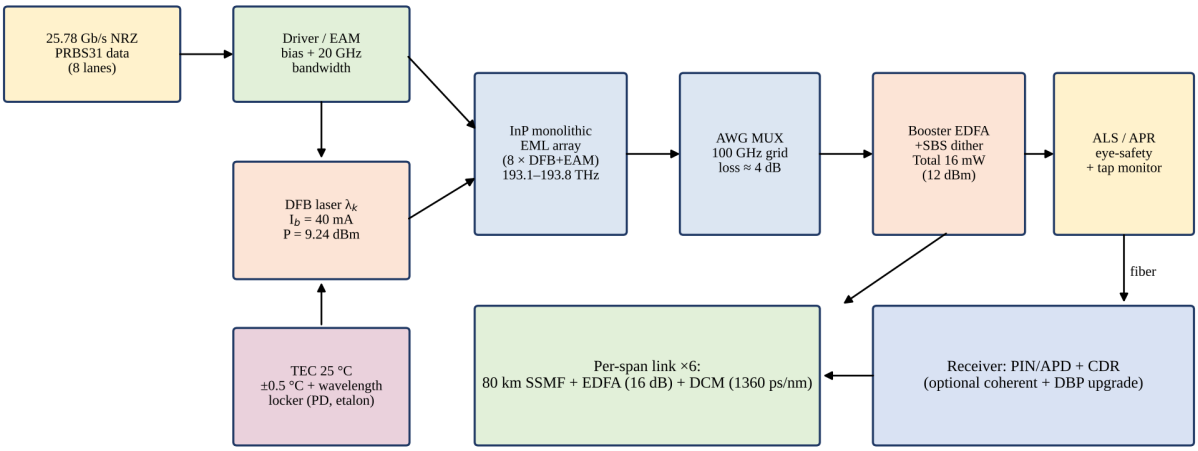
STAGE 6 – Verification Table

Requirement	Specified / Target	Achieved	Satisfactory?
Optical source selection	Single-mode source at 1550 nm, narrow line, η high	DFB; λ = 1550 nm; η_{int} = 80 %; SMSR > 35 dB	Yes
Modulation bandwidth and output power	≥ 17.5 GHz; ≥ 3 dBm/ch after losses	20 GHz (margin 2.5 GHz); 8.4 mW (9.24 dBm) CW	Yes (margin small)
Spectral/chirp and wavelength stability	±0.1 nm; low chirp	Drift 4.0 nm → ±0.05 nm with TEC ±0.5 °C; EAM low chirp	Yes
Nonlinear control and eye safety	φ_{NL} < ≈1 rad; P < P_{th}; safe	φ_{NL} = 0.33 rad; SBS 1.59 mW (dither); SRS 1.0 W; 16 mW total (ALS)	Yes, with SBS dither and ALS
Soliton / dispersion condition	N ≈ 1 for solitons	N = 0.17; P_0 = 68 mW; not feasible, NRZ + DCM	Soliton rejected (as intended)
Integration / DSP / standards strategy	G.957, IEEE 802.3bm/bs	InP EML array + AWG; DBP optional; standards test plan	Yes

The calculated and analyzed results are consistent with the original requirement: the weak points (SBS threshold of 1.59 mW and the 2.5 GHz bandwidth margin) have specific mitigations. Link OSNR check: OSNR = 58 + P_{ch}(dBm) − NF − loss − 10 log N = 58 + 3 − 5 − 16 − 7.8 ≈ **32 dB** (0.1 nm; NF = 5 dB assumed) – comfortably above the ≈ 17–20 dB typically needed for 25 Gb/s NRZ.

FINAL TRANSMITTER DESIGN – INTEGRATED ENGINEERING SOLUTION

Fig. 7 Block diagram of the proposed 8-channel DWDM transmitter



Parameter	Adopted value
Channels / grid	8 channels, 193.1–193.8 THz (1552.52–1546.92 nm), 100 GHz spacing
Laser type	DFB (InP) with external EAM (EML)
Nominal wavelength	1550 nm (E_g = 0.80 eV)
Bias current / threshold	40 mA / 12 mA (η_s = 0.3 mW/mA)
CW output power	8.4 mW (9.24 dBm); η_{int} = 80 %
Modulation bandwidth	20 GHz vs 17.5 GHz required (margin 2.5 GHz)
Data format	25.78125 Gb/s NRZ (64b/66b), PRBS31 test
TEC setting	25 °C set-point, ±0.5 °C stability, wavelength locker (±0.1 nm limit)
Launch / total power	2 mW (3 dBm) per channel / 16 mW (12 dBm) total
Power budget	9.24 dBm CW − ≈7 dB EAM/coupling − ≈4 dB AWG = ≈ −1.8 dBm/ch; booster EDFA ≈ +4.8 dB → 3 dBm/ch (assumed losses)
Nonlinear control	φ_{NL} = 0.33 rad; SBS dither; SRS 1.0 W; DCM 1360 ps/nm per span
Soliton	L_D = 11.25 km; N = 0.171; not feasible
Standards	ITU-T G.957/G.694.1, IEEE 802.3bm/bs, ANSI Z136 (ALS/APR)

How the choices work together: the DFB gives a narrow, single-mode line at 1550 nm with 8.4 mW; the EAM provides 25 Gb/s NRZ with low chirp so the laser can be held CW (stable wavelength, no relaxation-oscillation limit); the TEC plus locker keeps the 4 nm ambient drift inside ±0.1 nm so the 100 GHz grid is preserved; the modest 2 mW per channel keeps SPM at 0.33 rad and FWM/XPM/SRS negligible, while dither handles SBS; solitons are rejected (N = 0.17) and DBP is not needed; and InP integration with an AWG shrinks the module and the coupling loss.

FINAL ENGINEERING DECISION

WOULD YOU DEPLOY THE TRANSMITTER SOLUTION YOU CREATED? — YES (conditional).

Justification: (1) The source (DFB, 1550 nm, η_{int} 80 %, 8.4 mW) and the 20 GHz bandwidth (2.5 GHz margin) meet the 25 Gb/s requirement. (2) TEC + locker reduces the 4.0 nm drift to ±0.05 nm,

within the ±0.1 nm tolerance. (3) SPM (0.33 rad over six spans), XPM, FWM and SRS are small, and the SBS threshold (1.59 mW < 2 mW) is handled by external modulation and dither. (4) Soliton transmission (N = 0.17) and full DBP are correctly rejected for this IM-DD link. (5) The 16 mW aggregate output exceeds the assumed Class 1 limit and requires ALS/APR and labeling. (6) The InP monolithic transmitter can be verified against G.957 and IEEE 802.3bm/bs. Deployment is subject to bench confirmation of the SBS margin, per-span dispersion compensation, EAM bandwidth, and the eye-safety class.

Assumptions: D = 17 ps/(nm·km); $g_R = 6\times10^{-14}$ m/W; EDFA noise figure 5 dB; Class 1 limit ≈ 10 mW at 1550 nm; EAM insertion/coupling loss ≈ 7 dB; AWG loss ≈ 4 dB; typical component values in the comparison tables.